

How to automate result reporting

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<https://github.com/jimb0w/automation>

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Contents

1	Data cleaning	2
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1 Data cleaning

Document everything.

Say what you're going to do, do it, then comment on it.

I'm going to create a random health dataset with the following variables:

- Sex (0 = female; 1 = male)
- Age (in years)
- Diabetes status (0 = no diabetes; 1 = diabetes)
- Smoking status (0 = non-smoker; 1 = smoker)
- LDL-C (in mmol/L)
- Outcome (in outcome units)

As so:

```
cd /home/jimb0w/Downloads/tempauto/  
clear  
set obs 20000  
set seed 1312  
gen sex = runiform(0,1)  
gen age = rnormal(60,15)  
gen diabetes = runiformint(0,1)  
gen smoking = runiformint(0,1)  
gen ldl = rnormal(3,0.5)  
gen outcome = rnormal(200,30)+diabetes*rnormal(100,20)
```

Now I'm going to check the variables make sense:

```
. su(sex), detail
```

sex				
	Percentiles	Smallest		
1%	.0096901	.0000432		
5%	.0501992	.0001577		
10%	.1017735	.0001719	Obs	20,000
25%	.2508185	.0001787	Sum of wgt.	20,000
50%	.4990473		Mean	.4991676
		Largest	Std. dev.	.2875379
75%	.7493168	.9998247		
90%	.8966301	.9998952	Variance	.0826781
95%	.947588	.9999172	Skewness	.0015478
99%	.9893163	.9999394	Kurtosis	1.805996

```
. su(diab), detail
```

diabetes				
	Percentiles	Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	20,000
25%	0	0	Sum of wgt.	20,000
50%	0		Mean	.494
		Largest	Std. dev.	.4999765
75%	1	1		
90%	1	1	Variance	.2499765
95%	1	1	Skewness	.0240017
99%	1	1	Kurtosis	1.000576

```

. su(smoking), detail

```

smoking				
	Percentiles	Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	20,000
25%	0	0	Sum of wgt.	20,000
50%	1		Mean	.50085
		Largest	Std. dev.	.5000118
75%	1	1		
90%	1	1	Variance	.2500118
95%	1	1	Skewness	-.0034
99%	1	1	Kurtosis	1.000012

```

. su(age), detail

```

age				
	Percentiles	Smallest		
1%	24.66596	.835979		
5%	35.15182	1.354185		
10%	40.73132	4.960024	Obs	20,000
25%	49.76038	5.434331	Sum of wgt.	20,000
50%	59.77124		Mean	59.82388
		Largest	Std. dev.	14.99574
75%	69.96964	109.9906		
90%	78.95889	111.0537	Variance	224.8723
95%	84.55445	111.059	Skewness	-.0096071
99%	94.6009	115.5846	Kurtosis	3.032043

```

. su(ldl), detail

```

ldl				
	Percentiles	Smallest		
1%	1.81618	1.169115		
5%	2.166875	1.17411		
10%	2.354322	1.184871	Obs	20,000
25%	2.66082	1.20128	Sum of wgt.	20,000
50%	3.000187		Mean	2.997539
		Largest	Std. dev.	.50188
75%	3.334033	4.78299		
90%	3.636441	4.849288	Variance	.2518836
95%	3.817179	4.876576	Skewness	-.0133279
99%	4.168352	5.034541	Kurtosis	3.031805

```

. su(outcome), detail

```

outcome				
	Percentiles	Smallest		
1%	139.2732	87.21138		
5%	162.0865	91.83277		
10%	174.7943	91.93818	Obs	20,000
25%	199.8294	92.32668	Sum of wgt.	20,000
50%	244.0778		Mean	249.3994
		Largest	Std. dev.	59.80981
75%	299.1514	426.0753		
90%	329.6298	430.0988	Variance	3577.214
95%	345.9959	434.5064	Skewness	.163877
99%	374.6187	450.9283	Kurtosis	2.073516

It looks like these variables make sense and there don't appear to be any errors with the data. But we should check further: let's produce a histogram to check the continuous variables.

```

hist ldl
hist age
hist outcome
export delimited using randomhealthdata.csv, replace

```